

FRAME FOR PERFORMANCE

SSS 1-3 | Irrigation Engineering Garden

Frame and execute a research or engineering mission on hydraulic design and performance. Produce quantitative evidence that can support a design or management decision.

MISSION KIT flow tools, elevation/pipe data, calculator

EVIDENCE flow, head, loss, uniformity or efficiency evidence

GM-431 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Frame with flow tools, elevation/pipe data, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for flow, head, loss, uniformity or efficiency evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** No electrical or pressurized work without supervision. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Irrigation Engineering Garden, including a measurable acceptance criterion.

IRRIGATION RESEARCH MISSION

SSS 1-3 | Irrigation Engineering Garden

Frame and execute a research or engineering mission on hydraulic design and performance. Produce quantitative evidence that can support a design or management decision.

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EVIDENCE flow, head, loss, uniformity or efficiency evidence

GM-432 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Design with flow tools, elevation/pipe data, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for flow, head, loss, uniformity or efficiency evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
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AGRISHINE TASK

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INSTRUMENT THE DESIGN

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EVIDENCE flow, head, loss, uniformity or efficiency evidence

GM-433 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Instrument with flow tools, elevation/pipe data, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for flow, head, loss, uniformity or efficiency evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
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AGRISHINE TASK

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CALIBRATE FOR PERFORMANCE

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EVIDENCE flow, head, loss, uniformity or efficiency evidence

GM-434 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Calibrate with flow tools, elevation/pipe data, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for flow, head, loss, uniformity or efficiency evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** No electrical or pressurized work without supervision. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

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IRRIGATION RESEARCH MISSION

SSS 1-3 | Irrigation Engineering Garden

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MISSION KIT flow tools, elevation/pipe data, calculator

EVIDENCE flow, head, loss, uniformity or efficiency evidence

GM-435 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Model with flow tools, elevation/pipe data, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for flow, head, loss, uniformity or efficiency evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
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AGRISHINE TASK

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VALIDATE THE DESIGN

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EVIDENCE flow, head, loss, uniformity or efficiency evidence

GM-436 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Validate with flow tools, elevation/pipe data, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for flow, head, loss, uniformity or efficiency evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
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AGRISHINE TASK

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EVALUATE FOR PERFORMANCE

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GM-437 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Evaluate with flow tools, elevation/pipe data, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for flow, head, loss, uniformity or efficiency evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
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AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Irrigation Engineering Garden, including a measurable acceptance criterion.

IRRIGATION RESEARCH MISSION

SSS 1-3 | Irrigation Engineering Garden

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EVIDENCE flow, head, loss, uniformity or efficiency evidence

GM-438 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Optimize with flow tools, elevation/pipe data, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for flow, head, loss, uniformity or efficiency evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
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AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Irrigation Engineering Garden, including a measurable acceptance criterion.

STRESS-TEST THE DESIGN

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EVIDENCE flow, head, loss, uniformity or efficiency evidence

GM-439 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Stress-test with flow tools, elevation/pipe data, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for flow, head, loss, uniformity or efficiency evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
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AGRISHINE TASK

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RECOMMEND FOR PERFORMANCE

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Frame and execute a research or engineering mission on hydraulic design and performance. Produce quantitative evidence that can support a design or management decision.

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EVIDENCE flow, head, loss, uniformity or efficiency evidence

GM-440 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Recommend with flow tools, elevation/pipe data, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for flow, head, loss, uniformity or efficiency evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** No electrical or pressurized work without supervision. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Irrigation Engineering Garden, including a measurable acceptance criterion.